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(3)

# Measures of Central Tendency

# Measures of Central Tendency (The Geometric Mean)

***WHY DO WE NEED A  
GEOMETRIC MEAN?????***

## Measures of Central Tendency (The Geometric Mean)

**Example 3:-** The following table refers to the number of registered mortality of babies under 5 years of age in Ethiopia for the last 5 years. Find the average rate of the registered mortality of babies under 5 years of age through the given years using the AM and GM and interpret your result.

Measures of Central Tendency (The Geometric Mean)

| Years<br>(in Ethiopian Calendar) | Number of mortality |
|----------------------------------|---------------------|
| 2013                             | 30,000              |
| 2014                             | 45,000              |
| 2015                             | 40,500              |
| 2016                             | 50,625              |
| 2017                             | 70,875              |
|                                  |                     |

## Measures of Central Tendency (The Geometric Mean)

| Years (in Ethiopian Calendar) | Number of mortality | Rate of the previous year      | Percentage rate of the previous year |
|-------------------------------|---------------------|--------------------------------|--------------------------------------|
| 2013                          | 30,000              | -                              |                                      |
| 2014                          | 45,000              | $\frac{45,000}{30,000} = 1.5$  | $(1.5 - 1) \times 100\% = 50\%$      |
| 2015                          | 40,500              | $\frac{40,500}{45,000} = 0.9$  | $(0.9 - 1) \times 100\% = -10\%$     |
| 2016                          | 50,625              | $\frac{50,625}{40,500} = 1.25$ | $(1.25 - 1) \times 100\% = 25\%$     |
| 2017                          | 70,875              | $\frac{70,875}{50,625} = 1.4$  | $(1.4 - 1) \times 100\% = 40\%$      |

# Measures of Central Tendency (The Geometric Mean)

| Years<br>(in<br>Ethiopian<br>Calendar) | Number of<br>mortality | Average using<br>Arithmetic<br>Mean | Average using<br>Geometric<br>Mean |
|----------------------------------------|------------------------|-------------------------------------|------------------------------------|
| 2013                                   | 30,000                 | -                                   | -                                  |
| 2014                                   | 45,000                 | 1.5                                 | 1.5                                |
| 2015                                   | 40,500                 | 0.9                                 | 0.9                                |
| 2016                                   | 50,625                 | 1.25                                | 1.25                               |
| 2017                                   | 70,875                 | 1.4                                 | 1.4                                |
|                                        |                        | 1.2625                              | 1.2398                             |

## Measures of Central Tendency (The Geometric Mean)

*Interpretation (using AM):-* The registered mortality of babies in Ethiopia for the last 5 years has averagely increased by  $(1.2625 - 1) \times 100\%$   
 $= 0.2625 \times 100\% = 26.25\%$ .

*Interpretation (using GM):-* The registered mortality of babies in Ethiopia for the last 5 years has averagely increased by  $(1.2398 - 1) \times 100\%$   
 $= 23.98\%$ .

# Measures of Central Tendency (The Geometric Mean)

Recall that the product of the rate by the data of the previous year results the actual data of the current year.

| Years<br>(in Ethiopian<br>Calendar) | Number of<br>mortality | Rate of<br>the<br>previous<br>year |                               |
|-------------------------------------|------------------------|------------------------------------|-------------------------------|
| 2013                                | 30,000                 | -                                  |                               |
| 2014                                | 45,000                 | 1.5                                | $1.5 \times 30,000 = 45,000$  |
| 2015                                | 40,500                 | 0.9                                | $0.9 \times 45,000 = 40,500$  |
| 2016                                | 50,625                 | 1.25                               | $1.25 \times 40,500 = 50,625$ |
| 2017                                | 70,875                 | 1.4                                | $1.4 \times 50,625 = 70,875$  |



# Measures of Central Tendency (The Geometric Mean)

In the same manner if we multiply the data of the previous year by the AM and GM we computed we will get the following table

| Years<br>(in Ethiopian<br>Calendar) | Number of<br>mortality | Average using Arithmetic<br>Mean    | Average using Geometric Mean         |
|-------------------------------------|------------------------|-------------------------------------|--------------------------------------|
| 2013                                | 30,000                 | -                                   |                                      |
| 2014                                | 45,000                 | $1.2625 \times 30,000 = 37,875$     | $1.2398 \times 30,000 = 37,194$      |
| 2015                                | 40,500                 | $1.2625 \times 37,875 = 47,817.19$  | $1.2398 \times 37,194 = 46,113.12$   |
| 2016                                | 50,625                 | $1.2625 \times 47,817.19 = 60369.2$ | $1.2398 \times 46,113.12 = 57171.05$ |
| 2017                                | 70,875                 | $1.2625 \times 60369.2 = 76,216.11$ | $1.2398 \times 57171.05 = 70,880.66$ |

Far from the original result

Very near to the original result

## Measures of Central Tendency (The Geometric Mean)

*That is why we say the geometric mean provides an accurate measure of the average percentage rate of a series of numbers through time.*

# Measures of Central Tendency (The Geometric Mean)

## *Merits:*

- It is rigidly defined.
- It is not affected by the extreme items in the series.
- It is based on all the observations of the series.
- It is suitable for measuring the relative changes.

## Measures of Central Tendency (The Geometric Mean)

- It is useful in measuring growth of population
- It is used in averaging the ratios, percentages and in determining the rate gradual increase and decrease.

# Measures of Central Tendency (The Geometric Mean)

## *Demerits:*

- It is difficult to compute.
- It is not easy to understand.
- If there are negative values or zeros in the series, it can not be computed.

# The Harmonic Mean

# Measures of Central Tendency (The Harmonic Mean)

**Definition:-** The harmonic mean of N numbers

$x_1, x_2, x_3, \dots, x_N$ , denoted by HM is found by the formula:

$$HM = \frac{N}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_N}}$$

**Example 4:-** Find the HM of 5, 6, 8, 9, 11.

**Solution4:-**

$$HM = \frac{5}{\frac{1}{5} + \frac{1}{6} + \frac{1}{8} + \frac{1}{9} + \frac{1}{11}} =$$
$$\frac{5}{4752 + 3960 + 2970 + 2640 + 2160}$$
$$23760$$

## Measures of Central Tendency (The Harmonic Mean)

$$= \frac{5}{\frac{16482}{23760}} = \frac{5 \times 23760}{16482} = \frac{118600}{16482} = 7.2079$$

Here also note that

$$(HM = 7.2079) < (GM = 7.5019) < (AM = 7.8)$$

And in general  $HM \leq GM \leq AM$ .

***WHEN DO WE APPLY HARMONIC  
MEAN?????***



## Measures of Central Tendency (The Harmonic Mean)

**Example 5:-** A man travels 200 km each on the three days with speeds of  $70 \frac{\text{km}}{\text{hr}}$ ,  $80 \frac{\text{km}}{\text{hr}}$  and  $90 \frac{\text{km}}{\text{hr}}$ . Find the average speed.

**Example 6:-** A man travels 5 hours each on the three days with speeds of 70 km/hr, 80 km/hr and 90 km/hr. Find the average speed.

# Measures of Central Tendency (The Harmonic Mean)

Note that Average speed =  
$$\frac{\text{Total distance travelled}}{\text{Total time elapsed}}$$

$$\text{i.e. Average speed} = \frac{\sum_{i=1}^3 d_i}{\sum_{i=1}^3 t_i} = \frac{d_1 + d_2 + d_3}{t_1 + t_2 + t_3}$$

$$\text{and Speed} = \frac{\text{distance}}{\text{time}}$$

**Solution5:-** Given  $d_1 = d_2 = d_3 = 200$  KM and

$$s_1 = 70 \frac{\text{km}}{\text{hr}}, s_2 = 80 \frac{\text{km}}{\text{hr}} \text{ and } s_3 = 90 \frac{\text{km}}{\text{hr}}$$

Required: Average speed

# Measures of Central Tendency (The Harmonic Mean)

Since  $s_i = \frac{d_i}{t_i}$ , it follows that  $t_i = \frac{d_i}{s_i}$ .

Thus in our case  $t_1 = \frac{d_1}{s_1}$ ,  $t_2 = \frac{d_2}{s_2}$  and  $t_3 = \frac{d_3}{s_3}$

$$\Rightarrow t_1 = \frac{200\text{km}}{70\text{km/hr}} = \frac{200\text{km} \times \text{hr}}{70\text{km}} = \frac{200}{70}\text{hr}$$

$$t_2 = \frac{200\text{km}}{80\text{km/hr}} = \frac{200\text{km} \times \text{hr}}{80\text{km}} = \frac{200}{80}\text{hr}$$

$$t_3 = \frac{200\text{km}}{90\text{km/hr}} = \frac{200\text{km} \times \text{hr}}{90\text{km}} = \frac{200}{90}\text{hr}$$

# Measures of Central Tendency (The Harmonic Mean)

$$\therefore \text{Since average speed} = \frac{\sum_{i=1}^3 d_i}{\sum_{i=1}^3 t_i} = \frac{d_1 + d_2 + d_3}{t_1 + t_2 + t_3}$$

$$\therefore \frac{200\text{km} + 200\text{km} + 200\text{km}}{\frac{200}{70}\text{hr} + \frac{200}{80}\text{hr} + \frac{200}{90}\text{hr}} = \frac{3(200)\text{km}}{200\left(\frac{1}{70} + \frac{1}{80} + \frac{1}{90}\right)\text{hr}} =$$

$$\frac{3}{\frac{1}{70} + \frac{1}{80} + \frac{1}{90}} \text{ km/hr} \text{ which is the idea of the harmonic mean.}$$

$$\therefore = \frac{3}{\frac{7200 + 6300 + 5600}{504000}} \text{ km/hr} = \frac{3}{\frac{19100}{504000}} \text{ km/hr} =$$

$$\frac{3 \times 504000 \text{ km}}{19100} \text{ hr}$$

$$\therefore = \frac{1512000 \text{ km}}{19100} \text{ hr} = 79.16$$

## Measures of Central Tendency (The Harmonic Mean)

**Solution6:-** Given  $t_1 = t_2 = t_3 = 5$  hrs and

$$s_1 = 70 \frac{\text{km}}{\text{hr}}, s_2 = 80 \frac{\text{km}}{\text{hr}} \text{ and } s_3 = 90 \frac{\text{km}}{\text{hr}}$$

Required: Average speed

Since  $s_i = \frac{d_i}{t_i}$ , it follows that  $d_i = t_i \times s_i$ .

Thus in our case  $d_1 = t_1 \times s_1$ ,  $d_2 = t_2 \times s_2$  and  $d_3 = t_3 \times s_3$

$$\text{i.e. } d_1 = 5 \text{ hrs} \times 70 \frac{\text{km}}{\text{hr}}, d_2 = 5 \text{ hrs} \times 80 \frac{\text{km}}{\text{hr}} \text{ and } d_3 = 5 \text{ hrs} \times 90 \frac{\text{km}}{\text{hr}}$$

# Measures of Central Tendency (The Harmonic Mean)

$$\begin{aligned}
 \text{Since average speed} &= \frac{\sum_{i=1}^3 d_i}{\sum_{i=1}^3 t_i} = \frac{d_1 + d_2 + d_3}{t_1 + t_2 + t_3} \\
 &= \frac{(5\text{hrs} \times 70 \text{ km/hr}) + (5\text{hrs} \times 80 \text{ km/hr}) + (5\text{hrs} \times 90 \text{ km/hr})}{5\text{hrs} + 5\text{hrs} + 5\text{hrs}} \\
 &= \frac{5\text{hrs}(70 \text{ km/hr} + 80 \text{ km/hr} + 90 \text{ km/hr})}{3(5\text{hrs})} \\
 &= \frac{70 \text{ km/hr} + 80 \text{ km/hr} + 90 \text{ km/hr}}{3} \text{ which is the idea of the}
 \end{aligned}$$

$$\text{arithmetic mean, where average speed} = \frac{2400 \text{ km/hr}}{3} = 80 \frac{\text{km}}{\text{hr}}$$

# Measures of Central Tendency (The Harmonic Mean)

## *Merits and demerits of Harmonic Mean:*

### *Merits:*

- It is rigidly defined.
- Like AM and GM, it is also based on all the observations of a series i.e. it cannot be calculated ignoring any item of a series.
- It is capable of further algebraic treatment.

## Measures of Central Tendency (The Harmonic Mean)

- It can be calculated even when a series contains any negative value.
- It is most appropriate average under conditions of wide variations among the items of a series since it gives larger weight to smaller items.
- It is extremely useful while averaging certain types of rates and ratios.



## Measures of Central Tendency (The Harmonic Mean)

### *Demerits:*

- It is difficult to understand and to compute.
- It cannot be computed when one of the values is 0.
- It is necessary to know all the items of a series before it can be calculated.
- It is usually a value which may not be a member of the given set of numbers.

# The Median

## *Measures of Central Tendency (The Median)*

- The median is the middle number in *a sorted, ascending or descending* list of numbers and can be more descriptive of that data set than the average.
- The median is sometimes used as opposed to the mean when there are *outliers* in the sequence that might skew the average of the values.

## *Measures of Central Tendency (The Median)*

- If there is an *odd amount* of numbers, the median value is the number that is *in the middle*, with the same amount of numbers below and above.
- If there is an *even amount* of numbers in the list, the middle pair must be determined, *added together, and divided by two to find the median value.*

# *Measures of Central Tendency (The Median)*

## *Understanding the Median*

- Median is the middle number in a sorted list of numbers. To determine the median value in a sequence of numbers, *the numbers must first be sorted, or arranged, in value order from lowest to highest or highest to lowest.* The median can be used to determine an approximate average, or *mean*, but is not to be confused with the actual mean.

## *Measures of Central Tendency (The Median)*

### Median Example

To find the median value in a list with an odd amount of numbers, one would find the number that is in the middle with an equal amount of numbers on either side of the median. To find the median, first arrange the numbers in order, usually from lowest to highest.

## *Measures of Central Tendency (The Median)*

For example, in a data set of 3, 13, 2, 34, 11, 26, 47

(here  $N = 7$ ), the sorted order becomes:

2, 3, 11, 13, 26, 34, 47. The median is the number in the middle:

2, 3, 11, **13**, 26, 34, 47, which in this instance is 13 since there are three numbers on either side.

## *Measures of Central Tendency (The Median)*

- To find the median value in a list with an **even** amount of numbers, one must determine the *middle pair*, add them, and divide by two. Again, arrange the numbers in order from lowest to highest.



## *Measures of Central Tendency (The Median)*

For example, in a data set of 3, 13, 2, 34, 11, 17, 27, 47 (here  $N = 8$ ), the sorted order becomes:

2, 3, 11, 13, 17, 27, 34, 47. The median is the average of the two numbers in the middle:

2, 3, 11, **13, 17**, 26 34, 47, which in this case is fifteen

$$\frac{13 + 17}{2} = \frac{30}{2} = 15.$$

**Notation:-** The median is denoted by  $\tilde{x}$ .

## *Measures of Central Tendency (The Median)*

- In the case of *ungrouped frequency distribution*

to compute the median we use the idea of

cumulative frequency (*of less than type*) to arrange

the given data in order. Then after we use the formula:

$$\tilde{x}_{pos} = \left(\frac{N}{2}\right)^{th} h, \text{ when } N \text{ is even}$$

$$\text{and } \tilde{x}_{pos} = \left(\frac{N+1}{2}\right)^{th} h, \text{ when } N \text{ is odd (this is just}$$

to avoid the decimal)

## *Measures of Central Tendency (The Median)*

**Example 1:-** The following table refers to the weight (to the nearest Kg) of a sample of students at HiLCoE.

Find the median weight and interpret your result.

| Weight (in Kg) | Number of students | Cum. Frequency (Less than type) |
|----------------|--------------------|---------------------------------|
| 58             | 44                 |                                 |
| 59             | 136                |                                 |
| 60             | 270                |                                 |
| 61             | 443                |                                 |
| 62             | 211                |                                 |
| 63             | 101                |                                 |

## *Measures of Central Tendency (The Median)*

| Weight (in Kg) | Number of students | Cum. Frequency (Less than type) |
|----------------|--------------------|---------------------------------|
| 58             | 44                 | 44                              |
| 59             | 136                | 180                             |
| 60             | 270                | 450                             |
| 61             | 443                | 893                             |
| 62             | 211                | 1104                            |
| 63             | 101                | <b>1205 (ODD)</b>               |

*Median Class*

$$\text{Thus } \tilde{x}_{pos} = \left( \frac{N+1}{2} \right)^{th} = \left( \frac{1205+1}{2} \right)^{th} = \left( \frac{1206}{2} \right)^{th} = (603)^{rd}$$

## *Measures of Central Tendency (The Median)*

The  $603^{rd}$  student is in the 4<sup>th</sup> class and the weight of students in that class is 61 KG. *i.e.*

$$\tilde{x}_{val} = 61\text{kg}.$$

**Interpretation:-** About 50% of the students in the sample weigh 61 Kg or less (61 Kg or more).

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Any Questions



Any Questions

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Thank  
you